

EDUCATION

University of Toronto | Toronto, Canada

Aug 2025 - July 2026

Master of Mathematical Finance

Coursework: Pricing Theory, Machine Learning for Finance, Risk Management

Columbia University | New York City, United States

Aug 2015 - Feb 2017

Master of Science in Mechanical Engineering, Robotics Track

Coursework: Analysis of Algorithms, Numerical Methods

Tianjin Polytechnic University | Tianjin, China

Sep 2011 - Jun 2015

B.Sc. in Measurement Control Technology and Instruments

RESEARCH INTERESTS

- Asset Pricing
- Market Microstructure
- Optimization
- Computational Finance

RESEARCH

Design a Real-Time Dynamic Systems with convex quadratic programs optimization

Personal Open Source Research Project | 2023-2025

- Created and developed an efficient real-time Model Predictive Control (MPC) framework using Quadratic Programming (QP) for a floating base robotic system, prioritizing both computational efficiency and robustness
- Implemented C++ as the backend for QP optimization and integrated Python at the front end to enhance user-friendliness, utilizing Cython for this purpose.
- Open sourced and documented on GitHub <https://github.com/ke-zhang-rd/nullspace>

Thematic ETF Trading Pressure and Price Reversal Analysis

Courses Research Project | 2025-Present, Supervised by Prof. Hamid Arian, University of Toronto

- Analyzed whether idiosyncratic, actively managed weight increases in concentrated thematic ETFs generate temporary, non-fundamental price inflation and subsequent medium-term reversals
- Built a Propensity Score Matching (PSM) framework to construct a comparable control group of unaffected securities

KE ZHANG
kezhangdnl@gmail.com

Asymmetric Tail Risk Amplification from Passive Ownership

Courses Research Project | 2025-Present, Supervised by Prof. Hamid Arian, University of Toronto

- Examined whether higher passive ownership in U.S. equities disproportionately amplifies downside (left-tail) risk during periods of elevated market stress
- Implemented a panel regression with stock and time fixed effects, incorporating an interaction term between passive ownership and market-stress measures using high-frequency equities and options data

WORK EXPERIENCE

WEILAN Technology | Nanjing, China

2020 - 2023

Chief Engineer

- Led the innovative engineering team on humanoid and quadruped robot development
- Executed data-driven strategies that encompassed both data visualization and analysis

Brookhaven National Laboratory, U.S. Department of Energy | NY, USA

2017 - 2020

Computational Scientist in Data Acquisition, Management and Analysis Team (DAMA)

- Developed and refined algorithms and data-driven models specifically for image processing tasks.
- Analyzed data with machine learning and statistical modeling
- Supported COVID-19 vaccine development

TECHNICAL SKILLS

C++ • Python • Julia • CUDA • SQL • MongoDB • Git • Bloomberg • MATLAB • LaTeX

CERTIFICATION

CFA Level 1 passed (Nov. 2023), CFA Level 2 Candidate (May. 2026)

DIGITAL PORTFOLIO

Website: <https://ke-zhang-rd.github.io/kezhang>
LinkedIn: <https://www.linkedin.com/in/ke-zhang-36147a116/>
GitHub: <https://github.com/ke-zhang-rd>
Stackoverflow: <https://stackoverflow.com/users/10642305/ke-zhang>